INCH-POUND
MIL-DTL-0050869F (AR)
29 March 2004
USED IN LIEU OF
MIL-F-50869E (AR)
20 December 2003
MIL-F-48275(1)
29 October 1996

DETAIL SPECIFICATION

FUZE PIBD: M550, LESS SPITBACK ASSEMBLY

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-F-50869E and MIL-F-48275.

1. SCOPE

1.1 Scope. This specification covers the metal parts, assemblies, loading and assembling for one type of fuze designated as Fuze, PIBD: M550, Less Spitback Assembly.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval, of
MIL-A-8625	Anodic Coating for Aluminum and Aluminum Alloys
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-DTL-48669	Cartridge, 40MM: M550 Escapement Assembly
MIL-DTL-50872	Cartridge, 40MM: HEDP, M433

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DoD Preferred Method for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

US ARMY (ARDEC) DRAWINGS (see 6.3)

PRODUCT AND PACKING DRAWINGS

8864598 - Pin Firing
8864599 - Hammerweight Assembly
8864602 - Hammerweight Pin
8864603 - Firing Pin Spring
8864604 - Hammerweight Spring
8883760 - Actuator
8886350 - Ogive
8886355 - Body Assembly
8886356 - Body Fuze
8886357 - Escapement Assembly
8886360 - Fuze, PIBD, M550 Less Spitback
8886371 - Cartridge, 40MM: HEPD, M433
9266692 - Centerplate Assembly
9266693 - Centerplate
9276672 - O-ring
9287615 - Cartridge, 40mm, Test Vehicle with Fuze, M550
9287616 - Test Vehicle for Fuze M550

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9287617 - Cup and Skirt Inert Loading Assembly

INSPECTION EQUIPMENT DRAWINGS

8827889 - Gage, Receiver, Profile and Alignment, 40MM Cartridge
8827890 - Gage, Plug, Acceptance Check
8827895 - Limit Check

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM B117	Standard Method of Salt Spray (fog) Testing
ASTM E1255	Standard Practice for Radioscopy

(Copies may be requested online at www.astm.org or ordered from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Transportation vibration. Prior to transportation vibration test, the fuze assembly, shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After performance of the transportation vibration test the fuze assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature not in its proper position, damaged parts or foreign material and shall meet the requirements of functioning (3.6) and non-functioning (3.7).

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3.4 Jolt. The fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After being subjected to the jolt test, the fuze assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature not in its proper position, damaged parts or foreign material.

3.5 Jumble. The fuze assembly shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After being subjected to the jumble test, the fuze assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degrees from the safe position) or have any safety feature not in its proper position, damaged parts or foreign material.

3.6 Functioning. The fuze as a component of the higher assembly, the M433 Cartridge (Dwg. 8886371), shall meet all functioning requirements for that assembly.

3.7 Non-functioning. The fuze, as a component of the test vehicle (Dwg. 9287615), shall not function as evidenced by the spitback not functioning high order (premature burst).

3.8 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.9 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A – Not Applicable (N/A)
1 – Analysis
2 – Demonstration
3 – Examination
4 – Test

Classes of inspection
A – First article
B – Conformance

SECTION 3 REQUIREMENT	INSPECTION METHODS					INSPECTION CLASS		SECTION 4 INSPECTION METHOD
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4
3.3			X	X	X	X	X	4.5.13
3.4			X	X	X	X	X	4.5.11
3.5			X	X	X	X	X	4.5.12
3.6			X		X	X	X	4.5.14
3.7			X		X	X	X	4.5.15
3.8				X		X	X	4.5.21
3.9				X		X	X	4.5.22

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.3)
b. Conformance inspection (4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article inspection. When specified in the solicitation or contract, a sample of the fuze shall be subjected to first article inspection.

4.3.1 First article quantity. The contractor shall submit a first article for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Parts description	Drawings	Quantity
Pin Firing	8864598	25
Hammerweight pin	8864602	25
Firing Pin Spring	8864603	
-Before forming		25
-After forming		25
Hammerweight Spring	8864604	

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-Before forming		25
-After forming		25
Actuator	8883760	25
Ogive	8886350	25
Body Fuze	8886356	25
Centerplate	9266693	25
O-Ring	9276672	25
Hammerweight Assembly	8864599	50
Escapement Assembly	8886357	125
Body Assembly	8886355	25
Centerplate Assembly	9266692	50
Fuze Assembly	8886360	158

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination for defects of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and tests specified in Table II herein.

4.3.3 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE II. First Article Inspection

Fuze Assembly and Components

EXAMINATION OR TEST	NO. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Firing Pin Spring (Dwg. 8864603)			
Load Test	25	3.2	4.5.2
Body Fuze (Dwg. 8886356)			
Salt spray	5	3.2	4.5.3
Ogive (Dwg. 8886350)			
Salt spray	5	3.2	4.5.3
Centerplate (Dwg. 9266693)			
Deflection test	25	3.2	4.5.4
Hammerweight Assembly (Dwg. 8864599)			
Security of stake	50	3.2	4.5.5
Centerplate Assembly (Dwg. 9266692)			
Firing Pin retraction	125	3.2	4.5.9
Pull off test of hammerweight assembly	25 1/	3.2	4.5.7
Dislodging force test of hammerweight pins	25 1/	3.2	4.5.8
Retention check of firing pin	125	3.2	4.5.6
Fuze Assembly (Dw. 8886360)			
Air pressure of crimped joint	50	3.2	4.5.10
Jolt test	12	3.4	4.5.11, 4.5.16
Jumble test	96	3.5	4.5.12, 4.5.16
Transportation vibration	50	3.3	4.5.13, 4.5.16
Functioning	2/	3.6	4.5.14
Non-functioning	50 3/	3.7	4.5.15

NOTES:

1/ Hammerweight assemblies on each of the three (3) legs.

2/ This requirement is tested at higher assembly. See MIL-DTL-50872 (Cartridge, 40MM: HEDP, M433) for quantities required.

3/ After completion of the air pressure and transportation vibration test.

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of fuzes shall be produced with components complying with the following constraints:

a. Escapements from one interfix lot number, from one manufacturer.
b. Metal parts or components from one interfix lot number, from one manufacturer.

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4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Verification Level IV shall be used for all characteristics defined as Majors and Verification Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.11. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Fuze Body	SHEET 1 OF 1	8886356
			NEXT HIGHER ASSEMBLY 8886355

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Diameter setback spring cavity	Level IV	3.2	Gage
102.	Depth setback spring cavity	Level IV	3.2	Gage
103.	Diameter setback pin cavity	Level IV	3.2	Gage
104.	Depth setback pin cavity	Level IV	3.2	Gage
105.	Salt spray	1/	3.2	4.5.3
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-C-5541.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.2	Firing Pin Spring	SHEET 1 OF 1	8864603
			NEXT HIGHER ASSEMBLY 9266692

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Thickness	Level IV	3.2	Gage
102.	Spring Load Test	Level III	3.2	4.5.2
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22
NOTES:				

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.3	Firing Pin	SHEET 1 OF 1	8864598
			NEXT HIGHER ASSEMBLY 9266692

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Profile of firing pin point	Level IV	3.2	Gage
102.	Length from point to underside flange	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 8886350
4.4.2.4	Ogive			NEXT HIGHER ASSEMBLY 8886360
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR				
101.	Salt spray	1/	3.2	4.5.3
102.	Spherical radius	Level IV	3.2	Gage
MINOR				
201.	Color incorrect	Level II	3.2	Visual
202.	Protective finish with bare spot exposing base metal except rack marks	Level II	3.2	Visual
203.	Wall thickness	Level II	3.2	Gage
204.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-A-8625.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 9266693
4.4.2.5	Centerplate			NEXT HIGHER ASSEMBLY 9266692
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR 101.	Deflection test	32 1/	3.2	4.5.4
MINOR 201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Escapement Assembly	SHEET 1 OF 1	8886357
			NEXT HIGHER ASSEMBLY 8886355

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Rotor in armed or partially armed position 1/	100%	3.2	AAIE/4.5.1
2.	Setback spring missing 1/	100%	3.2	AAIE/4.5.1
3.	Setback pin missing or retracted 1/	100%	3.2	AAIE/4.5.1
4.	Detent not engaged in teeth of sector gear 1/	100%	3.2	AAIE/4.5.1
5.	Setback pin not blocking rotor 1/	100%	3.2	AAIE/4.5.1
6.	Detent spring missing 1/	100%	3.2	AAIE/4.5.1
7.	Detent missing 1/	100%	3.2	AAIE/4.5.1
8.	Pinion or verge assembly missing 1/	100%	3.2	AAIE/4.5.1
MAJOR	None defined			
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ For definitions see MIL-DTL-48669.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 8864599
4.4.2.7	Hammerweight Assembly			NEXT HIGHER ASSEMBLY 9266692
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None Defined			
MAJOR 101.	Security of Stake	32 1/	3.2	4.5.5
MINOR 201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.8	Centerplate Assembly	SHEET 1 OF 1	9266692
			NEXT HIGHER ASSEMBLY 8886355

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Retention check of firing pin	Level III	3.2	4.5.6
102.	Pull off test of hammerweight assembly	80 1/	3.2	4.5.7
103.	Dislodging force test of hammerweight pins	80 2/	3.2	4.5.8
104.	Firing pin retraction	Level III	3.2	4.5.9
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:

1/ Accept on zero (0) and reject on one (1).

2/ The lot shall be rejected if the average separation force of the entire sample is less than the specified average on the applicable drawing. If any one pin has a separation value less than the absolute minimum specified on the applicable drawing the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.9	Body Assembly	SHEET 1 OF 1	8886355
			NEXT HIGHER ASSEMBLY 8886360

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Crimp 360°	Level IV	3.2	Visual
102.	Max. height to top of centerplate	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22
NOTES:				

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.10	Fuze Assembly, Less Spitback	SHEET 1 OF 1	8886360
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Rotor in armed position	100%	3.2	AAIE/4.5.1
2.	Jolt 2/	12	3.4	4.5.11, 4.5.16
3.	Jumble 2/	96	3.5	4.5.12,4.5.16
4.	Non-functioning	50	3.7	4.5.15
5.	Transportation vibration	50	3.3	4.5.13, 4.5.16
MAJOR				
101.	Outside diameter	Level IV	3.2	Gage
102.	Air pressure test of crimped joint	Level III	3.2	4.5.10
103.	Functioning	1/	3.6	4.5.14
MINOR				
201.	Marking misleading or unidentifiable	Level II	3.2	Visual
202.	Improper crimp	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:

1/ The requirement is tested at higher assembly. See MIL-DTL-50872 for quantities.

2/ If a detonator initiates during this test, the failure shall be considered a “Major” defect if the fuze is in the unarmed position.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.11	Test Vehicle for M550 Fuze 1/	SHEET 1 OF 1	9287616
			NEXT HIGHER ASSEMBLY 9287615

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Depth to charge surface	Level IV	3.2	Gage
102.	Concentricity of charge with small inside diameter of skirt	Level IV	3.2	Gage
103.	Diameter of rotating bands	Level IV	3.2	Gage
104.	Security of test vehicle	Level III	3.2	4.5.18
105.	Chamber gage	100%	3.2	4.5.18/8827889, 8827890, 8827895
106.	Air pressure	Level III	3.2	4.5.20
107.	Specific gravity of inert material (Dwg. 9287617)	Level III	3.2	4.5.17
MINOR				
201.	Total length	Level II	3.2	Gage
202.	Perpendicularity of charge with small inside Dia of skirt	Level II	3.2	Gage
203.	Depth to bottom of cavity	Level II	3.2	Gage
204.	Marking misleading or unidentifiable	Level II	3.2	Visual
205.	Color of skirt incorrect	Level II	3.2	Visual
206.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.22

NOTES:
1/ All test vehicle components shall meet the Conformance Inspections slated in MIL-DTL-50872.

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4.5 Methods of inspection.

4.5.1 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (see 6.6).

4.5.2 Load requirement at each position. The applicable part shall be placed in an approved fixture and the load requirement at each position shall be measured. Observation shall be made for conformance with the applicable drawing requirement. This is a non destructive test. Parts so tested may be returned to the lot.

4.5.3 Salt Spray. The specified part shall be subjected to the salt spray test in accordance with ASTM B117. The duration of the test and the concentration of the salt is as specified on the respective drawings. This test is a destructive test. Parts so tested shall not be returned to the lot. Sampling and rejection criteria is as specified in MIL-A-8625.

4.5.4 Deflective test of centerplate. The centerplate shall be placed in an approved fixture and the minimum force specified on the applicable drawing shall be applied to the firing pinhead. The total deflection shall be measured. This is a destructive test. Parts so tested shall not be returned to the lot.

4.5.5 Security of stake of hammerweight assembly. The hammerweight assembly shall be placed in an approved fixture and the minimum force specified on the applicable drawing shall be applied. This is a destructive test. Parts so tested shall not be returned to the lot.

4.5.6 Retention check of firing pin spring on centerplate assembly. After firing pin spring (Dwg. 129444749) is assembled to firing pin (Dwg. 12044746) as shown on centerplate assembly (Dwg. 12944734), the centeplate assembly shall be placed in an approved fixture and only the three long firing pin spring legs shall support the weight of the centerplate assembly. During this test, the bottom of the centerplate (Dwg. 9266693) shall be suspended from surface of the fixture. This test is a non-destructive test. Parts so tested may be returned to the lot

4.5.7 Pull off test of the hammerweight assembly on centerplate assembly. The centerplate assembly shall be placed in an approved fixture and the minimum force specified on the applicable drawing shall be applied to the hammerweight pin at each of the three legs. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.8 Dislodging force test of hammerweight assembly pins on centerplate assembly. The centerplate assembly shall be placed in an approved fixture and the force necessary to push out the hammerweight pin at each of the three legs shall be measured. The force application orientation shall be in accordance with the applicable drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.5.9 Firing pin retraction of centerplate assembly. After completion of the retention check of firing pin spring on centerplate assembly of 4.5.6 the firing pin spring retraction shall be checked. The centerplate assembly shall be placed in an approved fixture. The fixture shall not compress the firing pin spring. The pre-load specified on the applicable drawing shall be applied to the firing pin representing the nominal spring force. The assembly shall then be rotated at the arming speed specified on the applicable drawing. The total deflection during rotation as well as the pre-load deflection shall be measured. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.10 Air pressure test of fuze crimped joint. The fuze shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the air pressure specified on the applicable drawing. The fixture and measured quantity of air shall be verified by demonstration through the detection of a known defective unit. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.11 Jolt. Prior to the jolt test the fuze assemblies shall be x-rayed in accordance with 4.5.16. The fuzes shall then be subjected to the test specified in Test A1-1 of MIL-STD-331B. After completion of the jolt test the fuzes shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.4. If one or more fuzes fails to meet the requirements of 3.4 or has experienced damage that compromises any safety feature of the fuze the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.12 Jumble. Prior to the jumble test the fuze assemblies shall be x-rayed in accordance with 4.5.16. The fuzes shall then be subjected to the test specified in Test A2-1.of MIL-STD-331B. After completion of the jumble test the fuzes shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.5. If one or more fuzes fails to meet the requirements of 3.5 or has experienced damage that compromises any safety feature of the fuze the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.13 Transportation vibration. Prior to transportation vibration test the fuze assemblies shall be x-rayed in accordance with 4.5.16. The fuzes shall then be packed in their regular shipping containers and be tested in accordance with MIL-STD-331 (Test No.B2, with vibration levels conforming to 6.1.2.1 of Test B1). The test shall be performed at ambient temperature only. Simulated fuzes may be used to insure a full box. After completion of the transportation vibration test the fuzes shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.3. If one or more fuzes fail to meet the requirements of 3.3 or has experienced damage that compromises any safety feature of the fuze the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. Parts subjected to subsequent tests shall not be returned to the lot. Only those fuze assemblies selected for non-functioning tests shall be subjected to this test.

4.5.14 Functioning. This test shall be conducted as a higher assembly requirement in accordance with specification MIL-DTL-50872 (Cartridge, 40MM: HEDP, M433).

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4.5.15 Non-functioning. After completion of the transportation vibration test the fuzes shall be assembled into cartridges with loaded spitback assemblies but with inert loaded cup and skirt assemblies in accordance with (Dwg. 9287615). The cartridges shall be fired from a M203 launcher against a vertical soft target, approved by the Contracting Officer. The target shall be placed forty-four (44) plus or minus two (2) foot from point of fire. The projectile velocity shall be measured and shall comply with the requirements of 4.5.15.1. Prior to FAT and each LAT, one test shot shall be fired at the proposed target to check for detonation at a range of 100 meters to proof test target. If one or more fuzes function on the target or a premature burst occurs the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. The subsequent function or non-function of the fuze down range of the target shall be measured.

4.5.15.1 Projectile Velocity. The projectile velocity shall be measured in accordance with the functional test procedures of MIL-DTL-50872. The velocity of the projectiles for this test shall be 226.5 to 267.5 feet/second. If more than 10% of tested rounds have velocities outside the required range, or if a velocity reading is not obtained, additional firings shall be made to replace the quantity over the 10% until acceptable velocities have been demonstrated.

4.5.16 X-ray of fuze assembly. Examination shall be performed with X-ray equipment or any other method acceptable to the contracting officer. Radioscopic examination equipment, procedures, and personnel shall be in accordance with the requirements of ASTM E1255 Annex A1. The fuze assembly shall be examined for compliance with 3.2, 3.3, 3.4, or 3.5 as applicable. The test shall be conducted in two (2) planes:

a. From the top of the fuze assembly down.
b. From the side

4.5.17 Specific gravity of inert material. The specific gravity of the inert material shall be determined as follows:

Weigh the empty cup and skirt assembly in grams and record as We. Fill the empty assembly with water at 75 degrees ± 2 degrees F., weigh in grams and record the weight as W_w. Determine the volume by subtracting W_e from W_e. Load the assembly with inert material. Surface of inert material to meet depth requirement, or liner assembly cavity may be formed or drilled. Weigh the assembly in grams and record the weight as W_1. Fill assembly with water at 75 degrees ±2 degrees F., weigh in grams and record the weight as W_b. Calculate the specific gravity as follows:

$$\text{Specific Gravity} = \frac{W_1 - W_e}{(W_w - W_e) - (W_b - W_1)}$$

Alternate method for specific gravity determination.

Prior to assembly, weigh empty body assembly with liner assembly and retainer in air (grams). Record the weight as “A”.

Prior to assembly, weigh the same empty body assembly with liner assembly and retainer submerged in water (grams). Record the weight as “B”.

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Dry the components and load with inert material using normal production methods.

Weigh the loaded assembly in air (grams). Record the weight as “C”.

Weigh the loaded assembly submerged in water (grams). Record the weight as “D”. Calculate the specific gravity of charge as follows:

$$\text{Specific Gravity} = \frac{C-A}{(C-A) - (D-B)}$$

Observation shall be made for conformance with the applicable drawing requirement.

4.5.18 Security of test vehicle. The test vehicle shall be placed in an approved fixture and the force specified on (Dwg. 9287616) shall be applied. Cartridge cases shall not be reused. Test vehicles may be reused.

4.5.19 Chamber gage failure. The test vehicle shall be placed in the chamber gage (Dwg. 8827889) and must meet the requirements specified on (Dwg. 9287616). This test is a non-destructive test, parts so tested may be returned to the lot.

4.5.20 Air pressure test of test vehicle. The test vehicle shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test, parts so tested may be returned to the lot.

4.5.21 Lot identification. Visually verify that each lot of fuzes is identified as the type, caliber, model and lot number in accordance with MIL-STD-1168.

4.5.22 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.8.

5. PACKAGING

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The components covered by this specification are intended for use with the Cartridge, 40MM: HEDP, M433.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.
c. ASSIST Online to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1).
d. Requirements for submission of first article sample.
e. Requirement for submission of design verification sample
f. Certificate of conformance for each lot or shipment of product.
g. Packaging requirements, if other than specified in Section 5.
h. Quality Conformance Inspection, if other than specified in Section 4 of this specification.
i. Provisions for critical characteristic controls.
j. Requirement for submission of inspection equipment designs.
k. Applicable National Stock number (NSN)
l. Serialization requirements, if applicable.
m. Government Furnished Material or Equipment.
n. Lists of drawings, publications and specifications, showing applicable revision dates.
o. Critical inspection equipment requirement (see 6.6).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.4 Definitions.

6.4.1 Premature burst. A premature burst is considered to have occurred if the round functions in the gun bore or prior to arming distance.

6.4.2 Early burst. A projectile that functions in flight further than the arming distance from the weapon muzzle.

6.5 Key Word Listing.

M433 Cartridge
M203 Grenade Launcher

6.6 Critical Inspection Equipment.

6.6.1 Automated acceptance inspection equipment. Equipment for which no, or minimal, human involvement is required in the acceptance determination. This would include equipment

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employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions. If readout is provided, the equipment will be categorized as automated even if operator interpretation is needed in the accept/reject decision.

6.6.2 System Reliability. The contractor should provide for each defect identified as Critical, a manufacturing and inspection system that assured no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.6.3 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristics of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and reinspected.

6.6.4 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in contract, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and the corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

6.6.5 Periodic verification. Periodic verification of the system error rate should be performed.

6.6.6 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.7 Revision/Amendments History. The following is a record of changes incorporated into this document:

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ECP	Description	Date Approved
R2S3021	Add a hole to top plate per attached sketch	09/25/2002
R2S3018	• Change test quantity identified in “Pin Retraction and Spring Removal Test” from 80 to 100 in paragraphs defining the test.	07/31/2002
R2S2021	• Implementation of new setback pin and spring for the M550 escapement used in the M550 fuze and M918 TP. • Clarification of the minimum arming time requirements in the item specifications.	07/01/2002
R7S3021	Arming time change with an approved inspection method	06/12/1997

6.8 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Army-AR
(Project 1390-2006-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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